

Supplemental information

**Conservation of antiviral systems across domains
of life reveals immune genes in humans**

Jean Cury, Matthieu Haudiquet, Veronica Hernandez Trejo, Ernest Mordret, Anael Hanouna, Maxime Rotival, Florian Tesson, Delphine Bonhomme, Gal Ofir, Lluís Quintana-Murci, Philippe Benaroch, Enzo Z. Poirier, and Aude Bernheim

Supplemental information

Conservation of antiviral systems across domains of life identifies immune genes in humans

In this file:

Figures S1-6

Additional Supplementary Materials:

Table S1, related to Figure 1. Human hits obtained with phylohmmer

Table S2, related to Figure 1. Human hits obtained with jackhammer

Table S3 related to Figures 3, 4, 5, 6. Scores of protein structures and co-folding predictions.

Table S4, related to Figure 1. List of DefenseFinder hits in eukaryotes after contamination removal

Table S5, related to Figure 2. List of human genes with GerpsRS and BS-REL scores

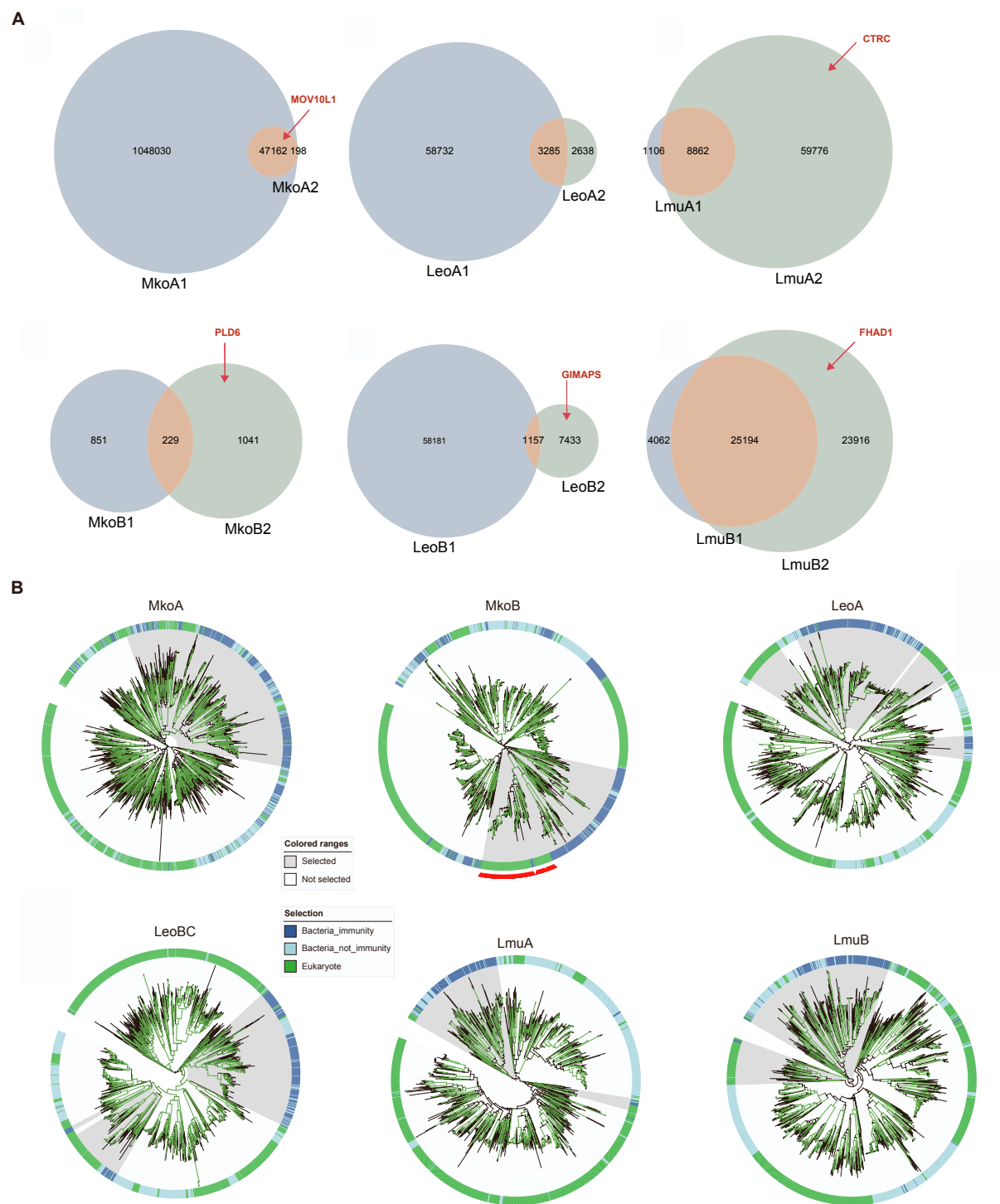


Figure S1. Homology analysis using prokaryotic and eukaryotic HMM profiles, related to Figure 1.

(A) Venn diagrams represent the intersection of hits obtained with prokaryotic HMM profiles (blue) and eukaryotic HMM profiles (green). Hits isolated by both analysis are highlighted in orange. Human genes explored in this study are indicated with a red arrow (all detected using eukaryotic HMM only except for MOV10L1, isolated in both analysis). **(B)** Phylogeny-driven selection of proteins for eukaryotic HMM profiles. Green branches correspond to ultra-fast bootstrap >95 (in IQTree). Ring colors indicate prokaryotic hits with colocalized antiphage partner (dark blue), prokaryotic hits without colocalized partners (light blue - hits not detected by DefenseFinder) and eukaryotic hits (green). Clades selected to build the HMM eukaryotic profile are indicated by a shade of gray.

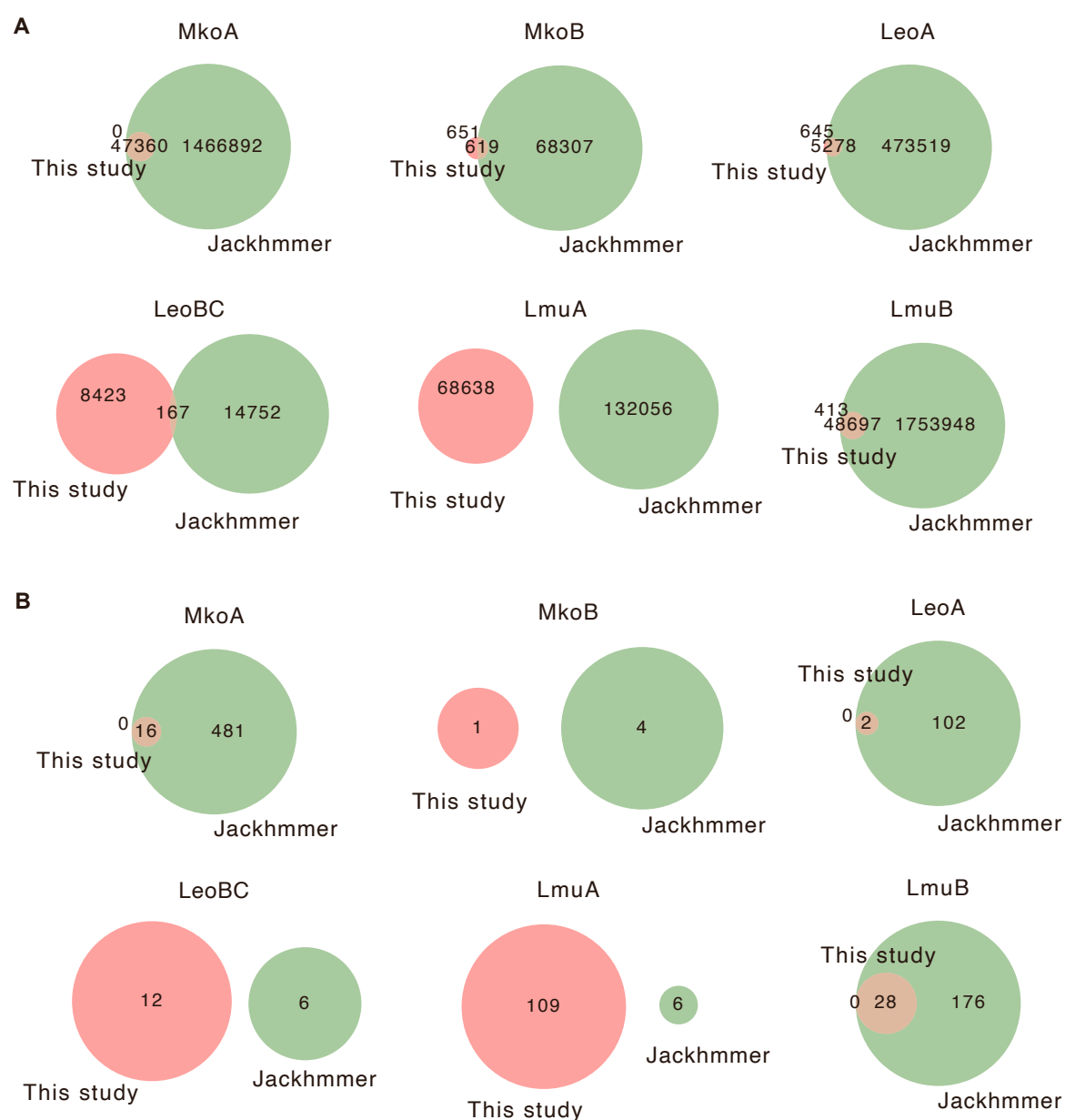


Figure S2, Eukaryotic and human hits obtained with our method and jackhmmmer, related to Figure 2.

Venn diagrams represent the intersection of hits obtained with jackhmmmer (green) and the in-house pipeline (red) in (A) all eukaryotic genomes or (B) in the human genome. Hits are listed in Tables S1 and S2.

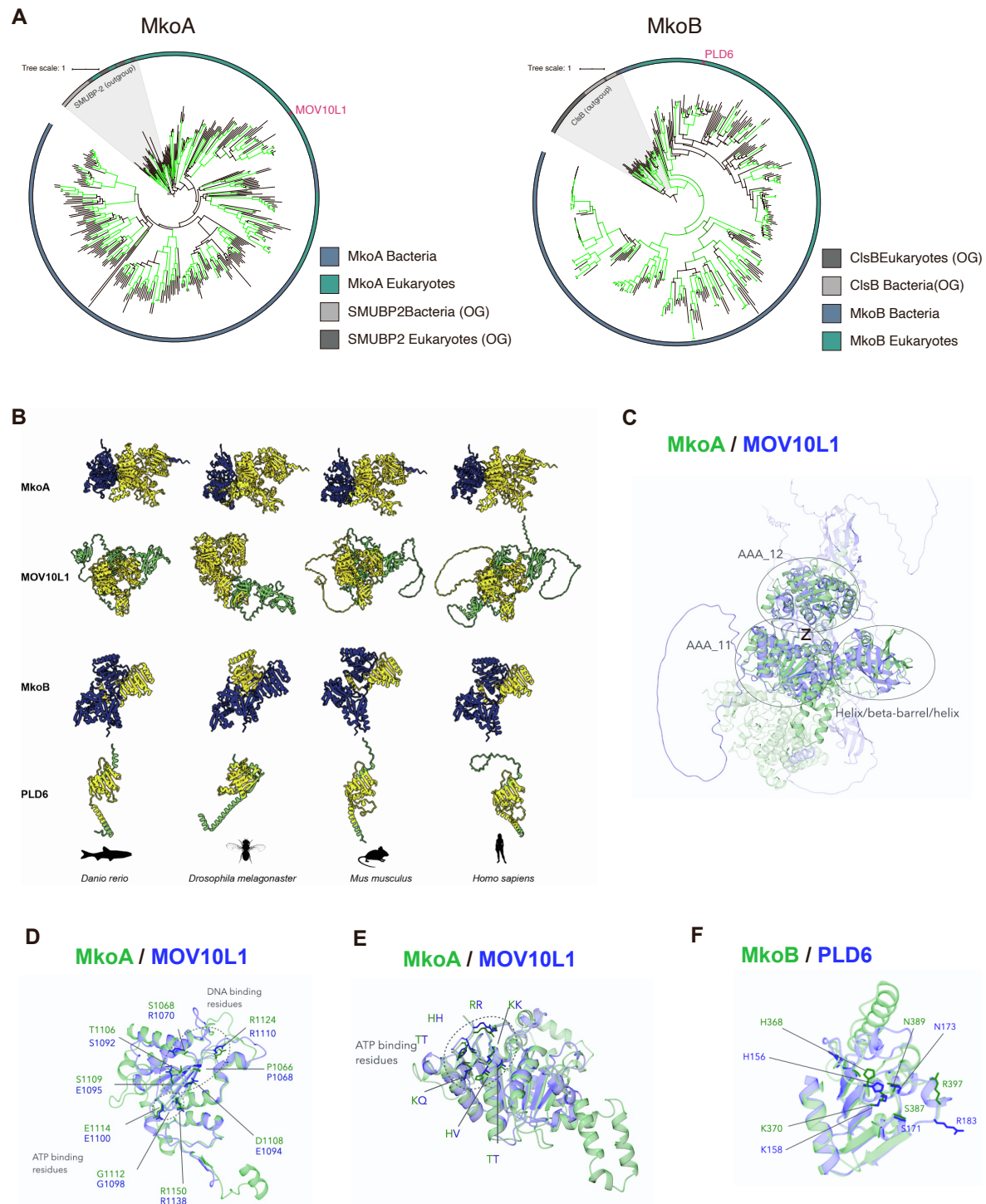


Figure S3. Phylogenies of MkoA, MkoB and structure comparison with MOV10L1, PLD6, related to Figure 3.

(A) Phylogenetic trees are rooted on the bacterial and eukaryotic outgroups (in shade of greys). Ultra-fast bootstraps >95 are highlighted in green. In the MkoA tree, the eukaryotic MkoA that are mixed with the outgroup, are in fact SMUBP-2 proteins, but were detected with the profile along other eukaryotic MkoA. **(B)** Structural comparisons of MOV10L1 (second row) and PLD6 (bottom row) of various eukaryotic species (in green) with the Mokosh proteins (in blue). For MOV10L1 (respectively PLD6) likely homologs, the yellow domain corresponds to the protein's optimal alignment to the MkoA (respectively MkoB) of Mokosh from *E. coli* ETEC H10407. Structures were predicted using AlphaFold. PDB identifiers and TM scores can be found in Table S3. **(C-F)** Prokaryotic proteins are depicted in green, and their eukaryotic homologs in blue. Portions of the proteins that do not align are represented with lower opacity. (C) Full structural alignment of MkoA (img_id: 2597791081) and MOV10L1 (uniprot:

Q9BXT6). Regions corresponding to the PFAM domains for AAA-11 and AAA-12 are highlighted. The helix/beta-barrel/helix region on the right does not show any sequence similarity despite displaying structural similarity. (D) Alignment of the AAA_12 domain of MkoA and MOV10L1. Key DNA binding and ATP binding positions are highlighted, with the nature of the residue and the position on the corresponding chain. (E) Same for the AAA_11 domain of MkoA and MOV10L1. (F) Alignment of PF13091 domains of MkoB (green) and PLD6 (blue), with key catalytic residues highlighted.

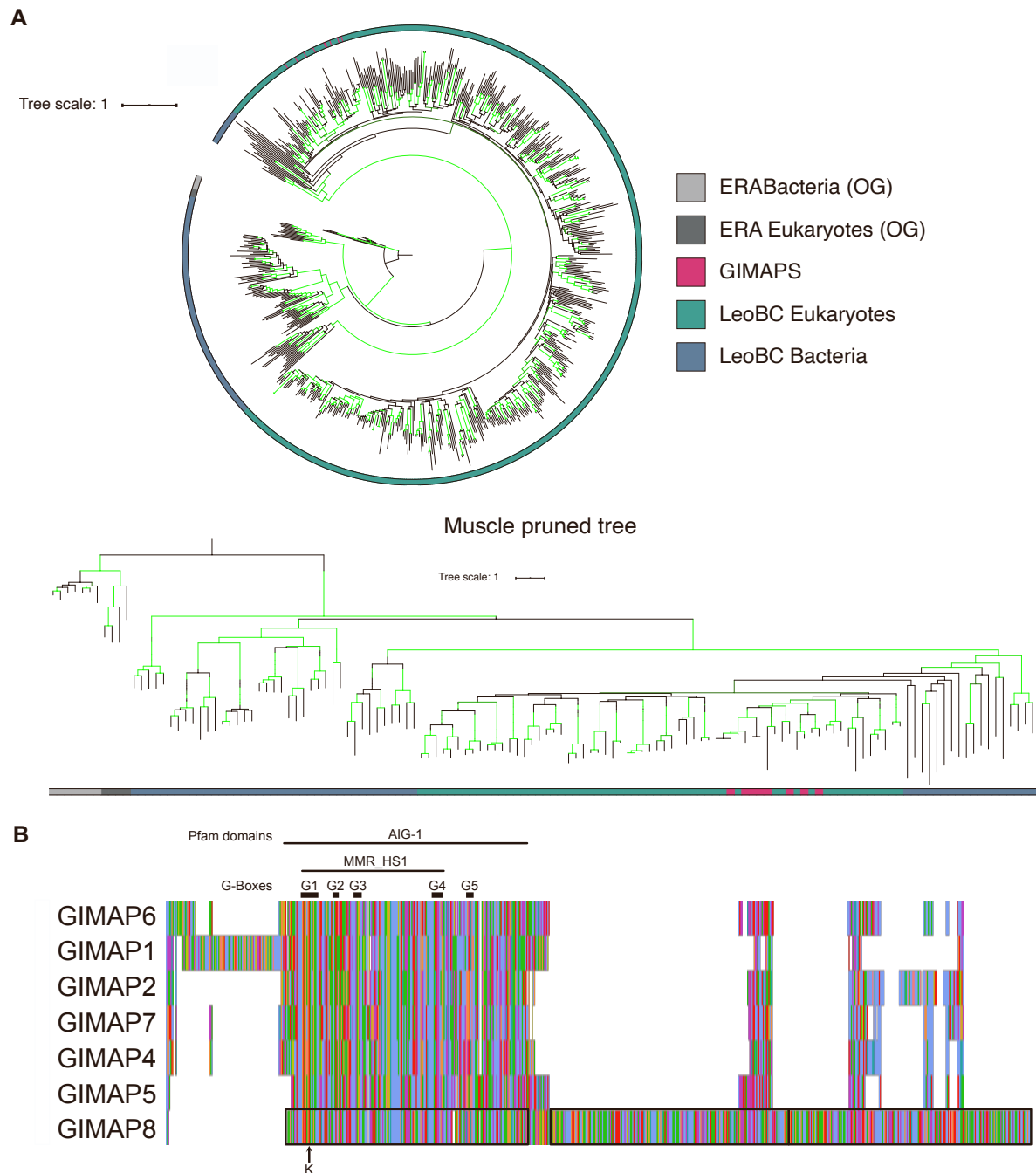


Figure S4. Phylogenies of LeoBC and GIMAPs, alignments of GIMAPs, related to Figure 4.

(A) Phylogenetic trees are rooted on the bacterial and eukaryotic outgroups (in shade of greys). Ultra-fast bootstraps >95 are highlighted in green. The bottom part is a pruned version of the above tree. **(B)** GIMAP8 is difficult to align properly with the other GIMAPs because it is composed of three AIG-1 domains (black rectangles) likely resulting from duplication-fusion events. To address this complexity, we initially excluded GIMAP8 from the set and performed the primary alignment using MUSCLE v5.1. We then used MAFFT v7.515 --add function, which is not available in MUSCLE, to incorporate GIMAP8 into the existing alignment. This method helps maintain the alignment integrity by accommodating GIMAP8's unique structure without compromising the overall alignment quality. Colors correspond to the ClustalX color scheme. Conserved G-boxes involved in GTP binding and hydrolysis are indicated. PFAM domains MMR_HS1 and AIG-1 (P-loop NTPase superfamily) are mapped on the sequences. Conserved lysine residue mutated in this study is indicated at the bottom.

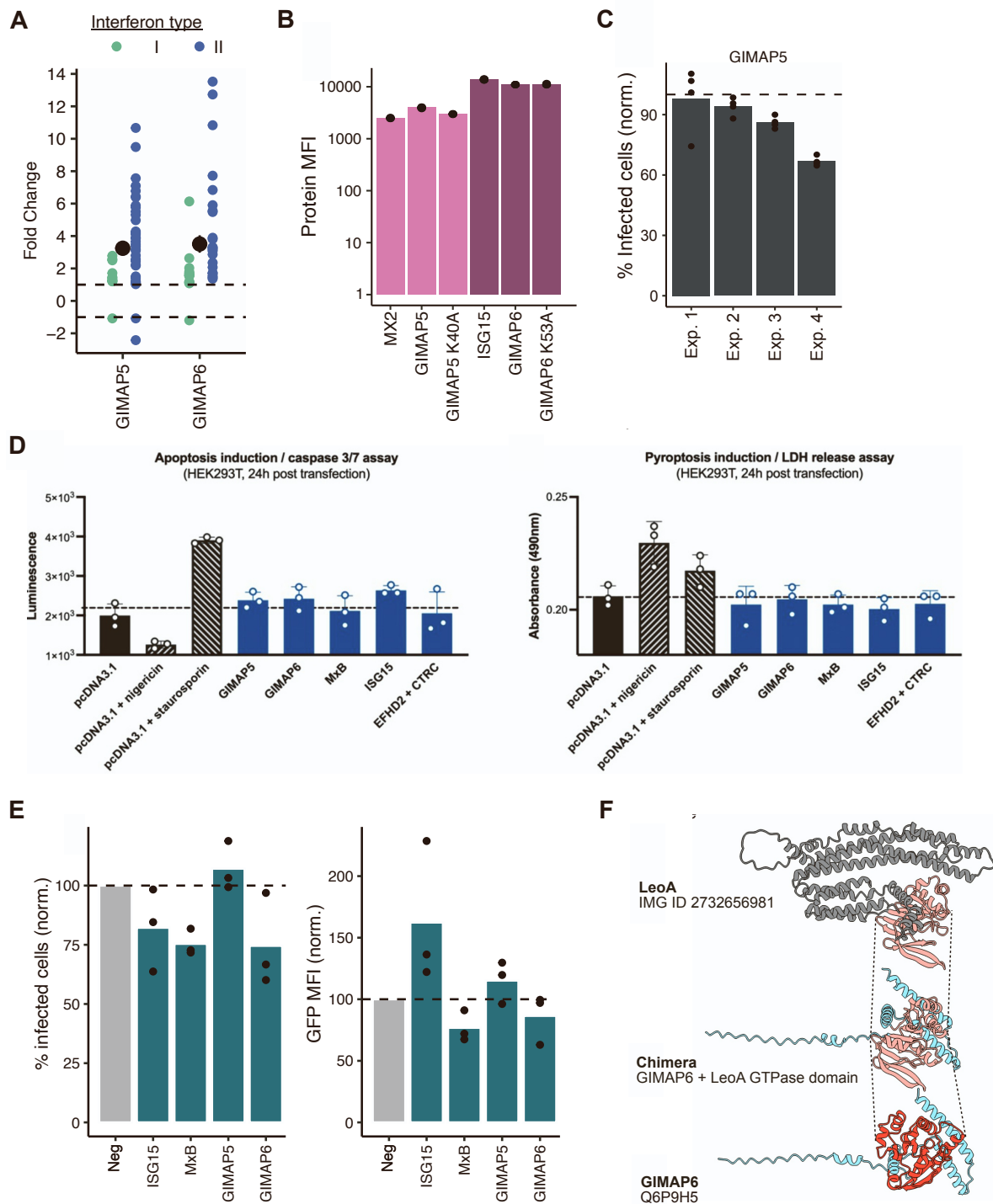


Figure S5. Human homologs of Eleos are antiviral proteins, related to Figure 5.

(A) Analysis of GIMAP5 and GIMAP6 expressions in tissue and cells after interferon treatment from the Interferome database (all values from the human datasets). Black points represent the global average.

(B) 293T cells are transfected with a plasmid coding for the indicated tagged protein. Intracellular mean fluorescence intensity (MFI) of each protein is monitored by intracellular immunostaining of the tag and flow cytometry. MxB, GIMAP5 and GIMAP5-K40A are FLAG-tagged, whilst ISG15, GIMAP6 and GIMAP6-K53A bear a HA-tag. Gates are adjusted to obtain similar protein expression, allowing meaningful comparison of infection between samples.

(C) Four independent experiments as shown in Fig. 3e, assessing the antiviral effect of GIMAP5 expression on HSV-1. $n = 4$ biological replicates. Paired t-tests with the value for cells expressing undetectable levels of protein, experiment 1 ($P=0.67$), experiment 2 ($P=0.07$), experiment 3 ($P=0.002$), experiment 4 ($P=0.0006$).

(D) 293T cells transfected with plasmids coding for GIMAP5, GIMAP6, MxB, ISG15, EFHD2 and CTRC were assessed for

apoptosis activation (Caspase 3/7 assay, left) and pyroptosis activation (LDH release assay, right). Staurosporin (resp. nigericin) is used as positive control for induction of apoptosis (resp. pyroptosis). **(E)** Infection of 293T-transfected cells with VSV is monitored by flow cytometry at 48h. Percentage of infected cells or GFP mean fluorescence intensity in cells expressing the indicated protein (x-axis) is normalised on the percentage of infected cells in cells with undetectable expression of the indicated protein (paired data from a single population of transfected cells). $n = 3$ biological replicates. All comparisons against control population (Neg.) yield $p > 0.05$ (Paired Student's t -test). **(F)** AlphaFold2 models of LeoA, GIMAP6, and the GTPase-domain chimera. In pink and red, GTPase domains (from G1 to G5 boxes), in grey and blue, the rest of the protein.

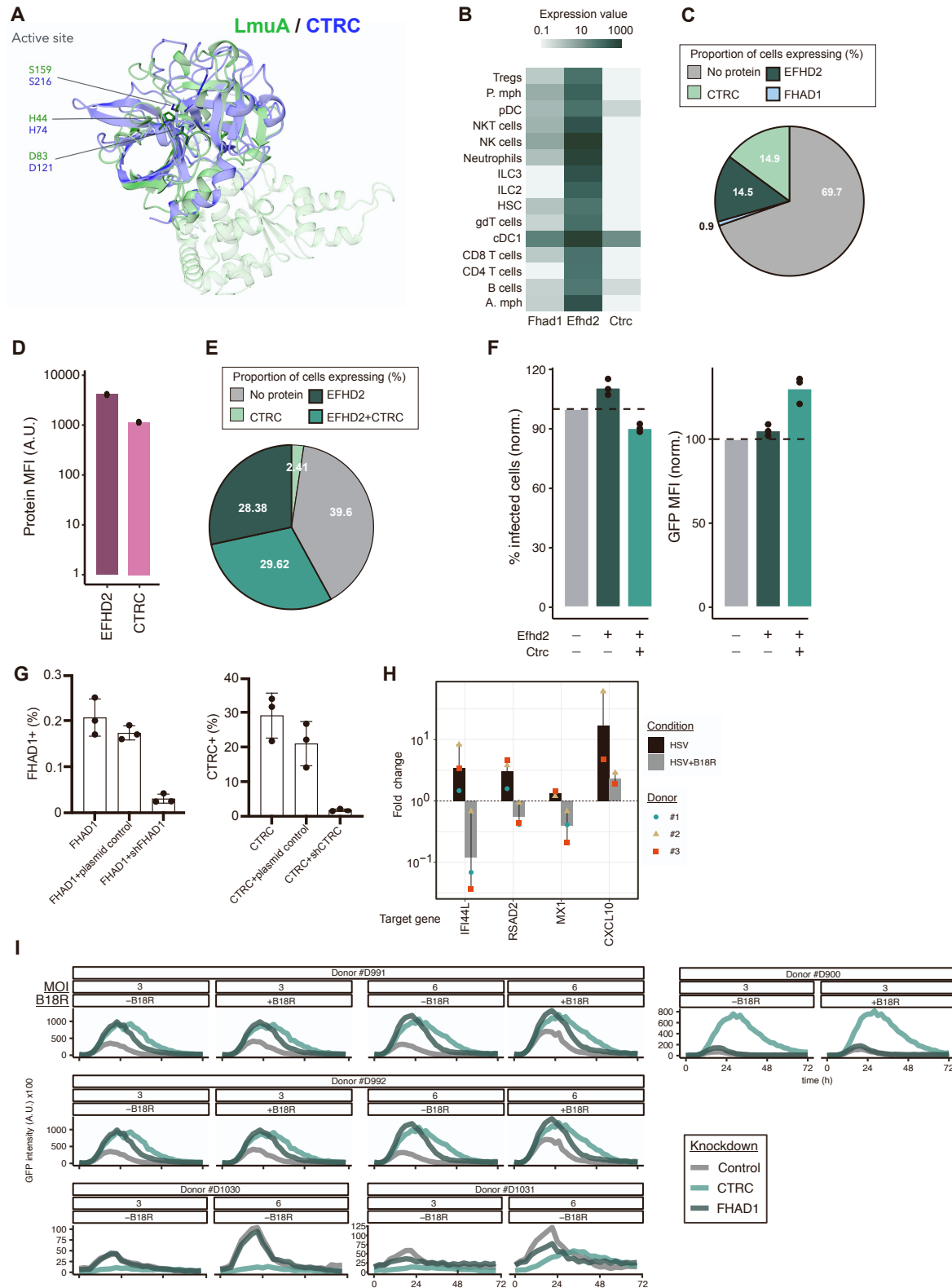


Figure S6. Human homologs of Lamassu are antiviral proteins, related to Figure 6.

(A) Full protein alignment of LmuA (ncbi id: EEV5712270.1, green) and CTRC (uniprot: Q99895, blue), aligned on the catalytic triad. (B) Expression of FHAD1, EFHD2 and CTRC in mouse immune cells (RNAseq data from the ImmGen Project, expression value normalised by DESeq2). (C) 293T cells are transfected with a mix of plasmids encoding for FHAD1, EFHD2 and CTRC and infected with HSV-1. Protein expression is quantified by flow cytometry at 48h post infection (average of $n = 4$ replicates). Pie

chart represents (for a single transfection) the percentage of cells expressing no protein, only FHAD1, EFHD2 or CTRC. **(D)** Intracellular mean fluorescence intensity (MFI) of each protein (in the CTRC+EFHD2+ population) is monitored by intracellular immunostaining targeting the tag followed by flow cytometry. EFHD2 is FLAG-tagged, whilst CTRC bears a HA-tag. **(E)** 293T cells are transfected with a mix of plasmids encoding for EFHD2 and CTRC and infected with HSV-1. Expression of CTRC and EFHD2 is quantified by flow cytometry at 48h post infection. Pie chart represents (for a single transfection) the percentage of cells expressing no protein, only CTRC or EFHD2, or both proteins (average of $n = 4$ biological replicates). **(F)** Infection of 293T-transfected cells with VSV is monitored by flow cytometry at 48h. Percentage of infected cells or GFP mean fluorescence intensity expressing the indicated protein (x-axis) is normalised on the percentage of infected cells in cells with undetectable expression of the indicated protein (paired data from a single population of transfected cells). $n = 3$ biological replicates. All comparisons against control population (Neg.) yield $p > 0.05$ (Paired Student's *t*-test). **(G)** 293T cells were transfected with plasmids encoding for FHAD1 or CTRC together with a control plasmid or with a plasmid coding for a shRNA targeting the FHAD1 or CTRC transcripts. Cells were collected at 6h post transfection, and expression of FHAD1 or CTRC was measured by immunostaining and flow cytometry. **(H)** Human primary macrophages derived from isolated blood monocytes were infected with HSV-1 in the presence or not of B18R. Transcriptional upregulation of ISGs (IFI44L, RSAD2, MX1) and pro-inflammatory cytokines were measured by RT-qPCR. Data is normalized on non-infected controls. **(I)** FHAD1 or CTRC levels are knock-down by shRNA transduction in blood-derived human myeloid cells, then infected with HSV-1 (MOI 3 or 6). Infection is monitored through GFP measurement by live video-microscopy (see Methods). Cells were treated with soluble interferon alpha/beta receptor B18 (+/- B18R) that blocks the type-I interferon response by quenching interferons in the media. Each blood donor is drawn to its own intensity scale, which cannot directly be compared between experiments.